COSC264
Introduction to Computer Networks and the Internet

Application Protocols: Introduction to Email and DNS

Dr Barry Wu

WRC
University of Canterbury
Barry.wu@canterbury.ac.nz

Learning Objectives

- At the end of the lectures, you will be able to:
  - Discuss the applications for emails
  - Explain that basic functionality of SMTP
  - Explain the basic function of mail access protocols, including POP3 and IMAP
  - Describe the key elements of MIME
  - Explain Internet domains and domain names
  - Discuss the operation of the Domain Name System

Outline

- Electronic Mail (Email)
  - SMTP, POP3, IMAP
  - MIME
- Domain Name System (DNS)

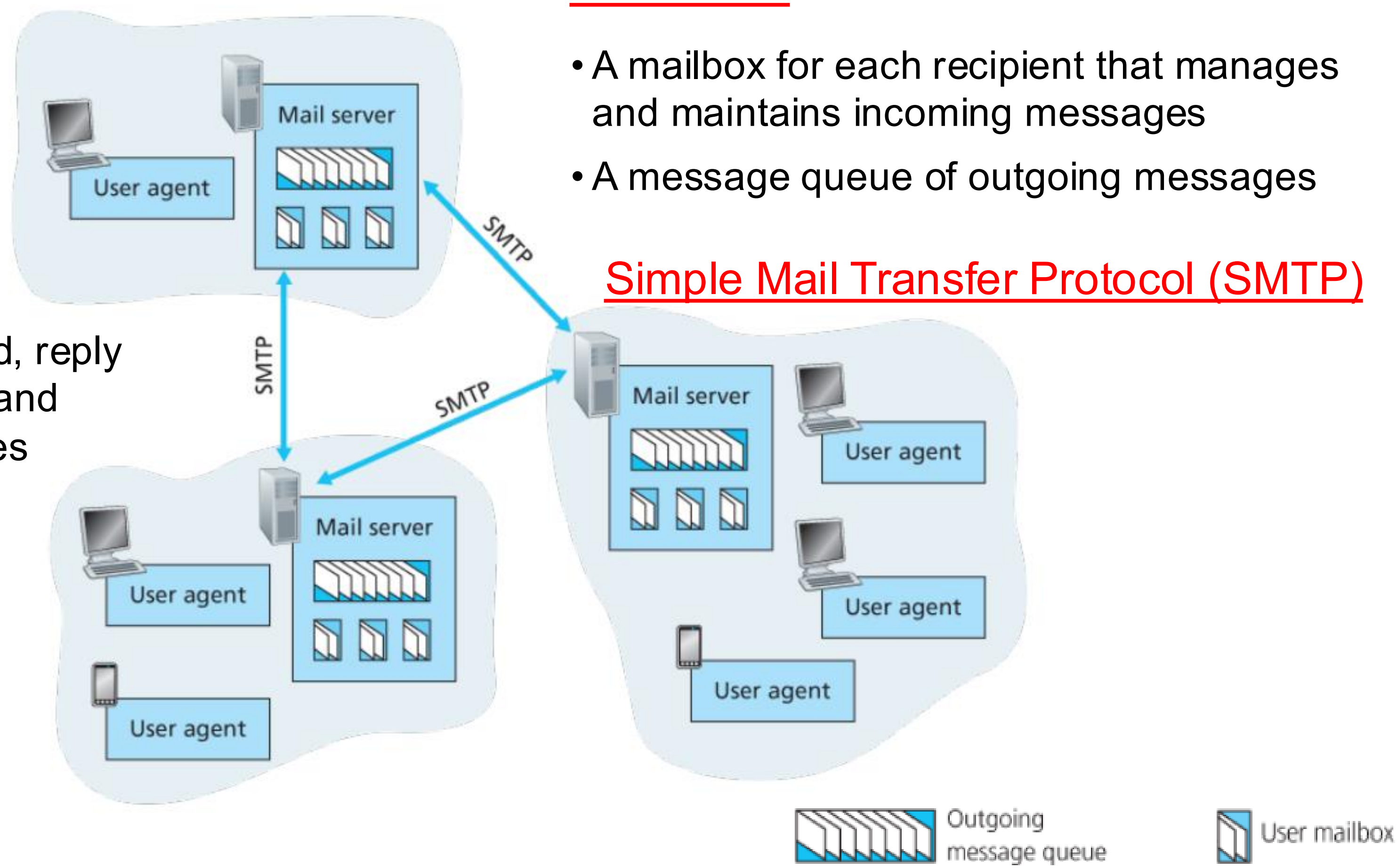
Email

Email Overview

- A high-level view of the Internet email system

User agent

- Allow users to read, reply to, forward, save, and compose messages
- Example: Outlook



Mail server

- A mailbox for each recipient that manages and maintains incoming messages
- A message queue of outgoing messages

Simple Mail Transfer Protocol (SMTP)

Source: James F. Kurose & Keith W. Ross, Computer networking: a top-down approach featuring the Internet, 3rd edition, 2005, Figure 2.15.

Simple Mail Transfer Protocol (SMTP)

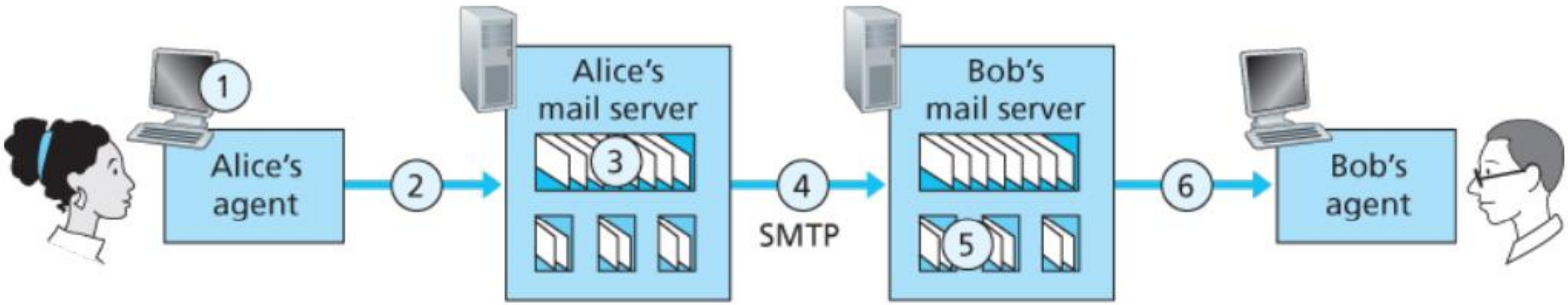
- SMTP is a mail transport and delivery protocol
  - Also used as a “mail submission” protocol
- SMTP has two sides
  - A client side executes on the sender’s mail server.
  - A server side executes on the recipient’s mail server.
- Both the client and server sides of SMTP run on every mail server
  - When a mail server sends mail to other mail servers, it acts as an SMTP client.
  - When a mail server receives mail from other mail servers, it acts as an SMTP server.

Basic Operation of SMTP

1. Alice uses her UA to compose the message

3. The SMTP client opens a TCP connection to an SMTP server on Bob's mail server

5. The SMTP server receives the message; Bob's mail server places the message in Bob's mailbox.



2. Alice's UA sends the message to her mail server

4. The SMTP client sends Alice's message over the TCP connection

6. Bob invokes his UA to read message

Source: James F. Kurose & Keith W. Ross, Computer networking: a top-down approach featuring the Internet, 3rd edition, 2005, Figure 2.16.

SMTP vs. HTTP

- Common characteristics
  - Transfer files from one host to another
  - Persistent connections
- Differences

HTTP	SMTP
Pull protocol	Push protocol
Synchronous communication	Asynchronous communication
Direct request-response model	Store-and-forward architecture
	based on 7-bits ASCII

An Example of Internet Message Format

- A message consists of header fields followed, optionally, by a body
- A message is ASCII text

Header lines

```
Date: October 8, 2009 2:15:49 PM EDT
From: "William Stallings" <ws@shore.net>
Subject: The Syntax in RFC 5322
To: Smith@Other-host.com
Cc: Jones@Yet-Another-Host.com
```

Body

```
Hello. This section begins the actual
message body, which is delimited from the
message heading by a blank line.
```

Multipurpose Internet Mail Extensions (MIME) for None-ASCII Data

- An extension to the RFC 5322
  - The specification is provided in RFCs 2045 through 2049

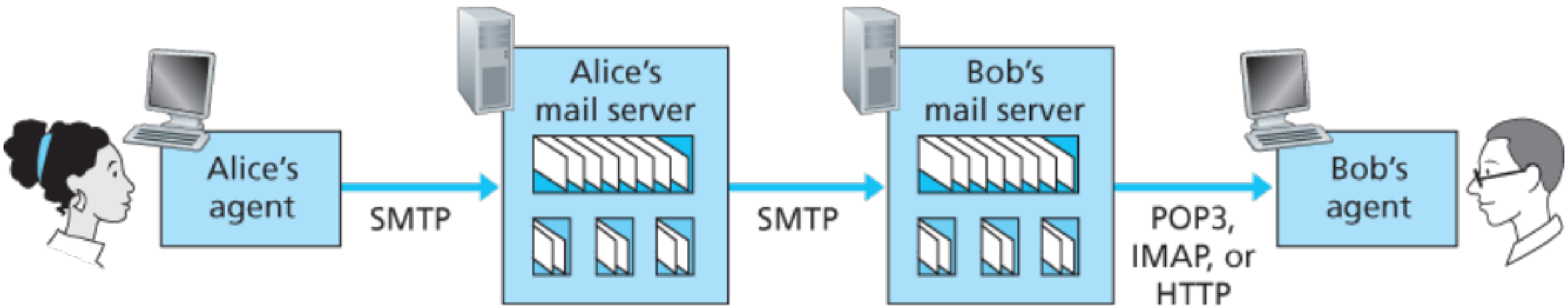
```
From: alice@crepes.fr
To: bob@hamburger.edu
Subject: Picture of yummy crepe.
MIME-Version: 1.0
Content-Transfer-Encoding: base64
Content-Type: image/jpeg

(base64 encoded data ......
......base64 encoded data)
```

Encode binary data as printable text

Mail Access Protocols

- Email retrieval from the recipient's mail server to the recipient's user agent
  - POP3: Post Office Protocol version 3 [RFC 1939]
  - IMAP: Internet Mail Access Protocol version 4 [RFC 3501]



HTTP for web-based email

Source: James F. Kurose & Keith W. Ross, Computer networking: a top-down approach featuring the Internet, 3rd edition, 2005, Figure 2.17.

POP3

Connection establishment	Authentication state	Transaction state	Update state
• The user agent establishes a TCP connection to the mail server on port 110.	• The user agent sends a username and a password (cleartext) to authenticate the user.	• The user agent retrieves messages. • The user agent can mark messages for deletion, remove marks, and obtain mail statistics.	• The mail server deletes the messages that were marked for deletion and closes the connection.

POP3 (cont.)

- What are some notable issues with POP3?
  - POP3 does not synchronise emails across multiple devices
  - POP3 does not support server-side folder management, so users can’t organise emails into folders on the server

IMAP

- IMAP keeps messages on the server and replicates copies to the clients.
- Users can organise messages in folders.
- IMAP maintains user state information across sessions.
- IMAP4 allows a user agent to obtain components of messages.

Web-based Email

Web-based email is accessed through a web browser

HTTP for communication between the user's browser and the mail server

Web-based email services

Outlook

Gmail

DNS

Domain Name System (DNS)

- The IP address provides a way of uniquely identifying devices on the Internet.
- The number scheme is effective for computer processing but is not convenient for users.

DNS Overview
[RFC 1034, 1035]

A directory lookup service

- Provide a mapping between the name of a host on the Internet and its numerical address

A distributed database

- Implemented in a hierarchy of DNS servers

An application-layer protocol (over UDP on port 53)

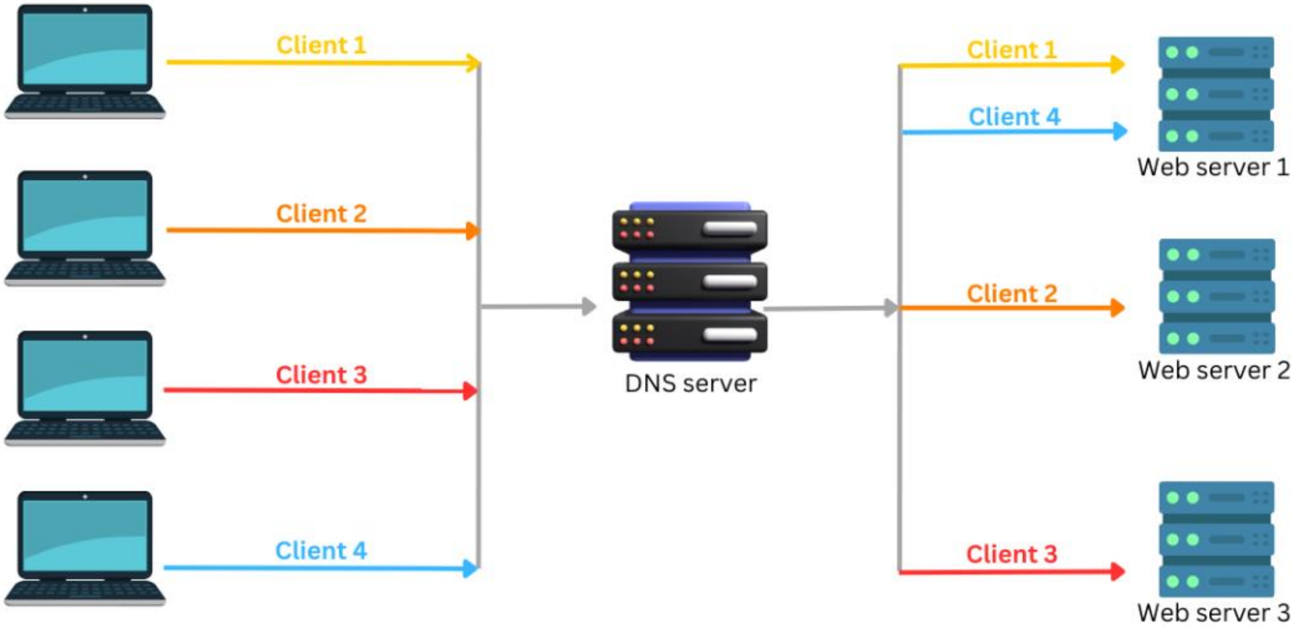
- Allow hosts to query the distributed database

DNS Services

- **Hostname to IP address translation**
  - Commonly employed by HTTP, SMTP, etc.
- **Host aliasing**
  - Canonical and alias names
  - Example: *relay1.west-coast.enterprise.com* could have two aliases such as *enterprise.com* and *www.enterprise.com*
- **Mail server aliasing**
  - Example: Bob's email address might be *bob@outlook.com*; the canonical hostname might be *relay1.west-coast.outlook.com*

DNS Services (cont.)

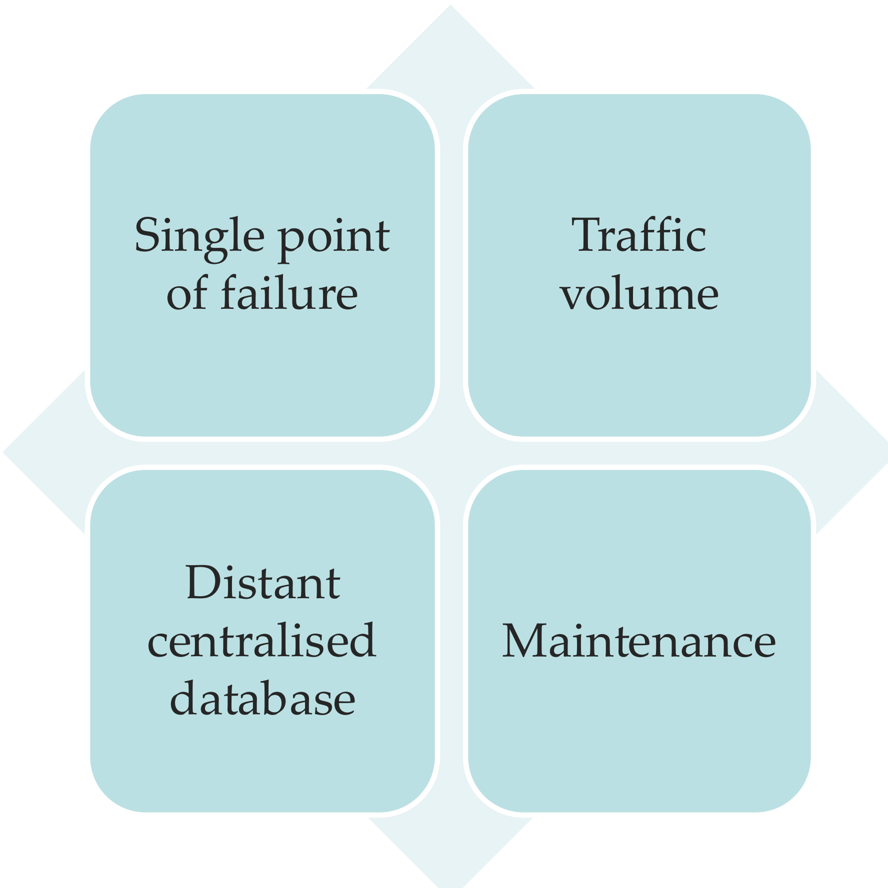
- **Load distribution**
  - Perform load distribution among replicated servers
  - Example: round-robin DNS

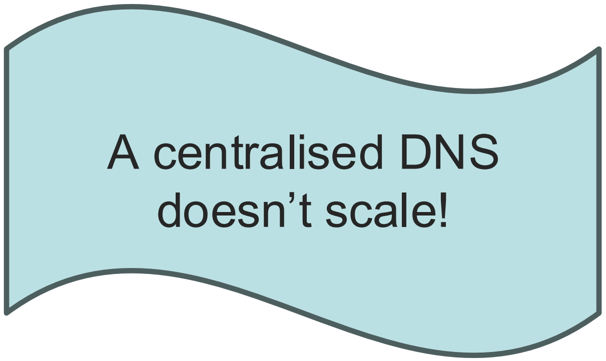


Source: https://www.cloudns.net/wiki/article/182/

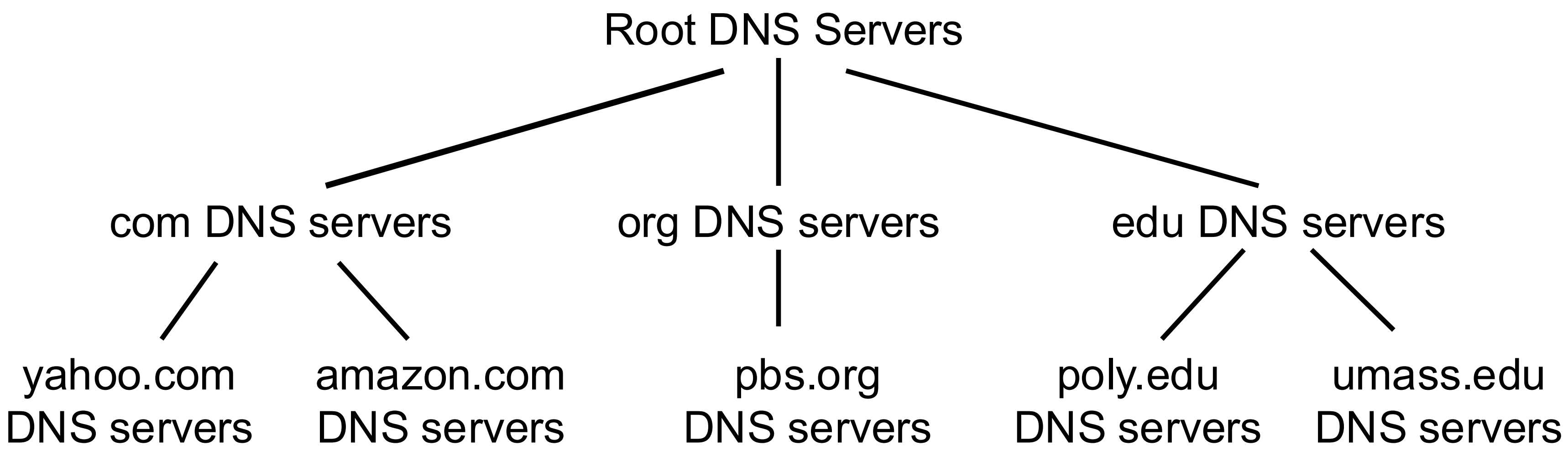
Why not centralised DNS?

Problems with a centralised design





A Distributed, Hierarchical Database



Source: https://www.iana.org/domains/root/servers

In-Confidence

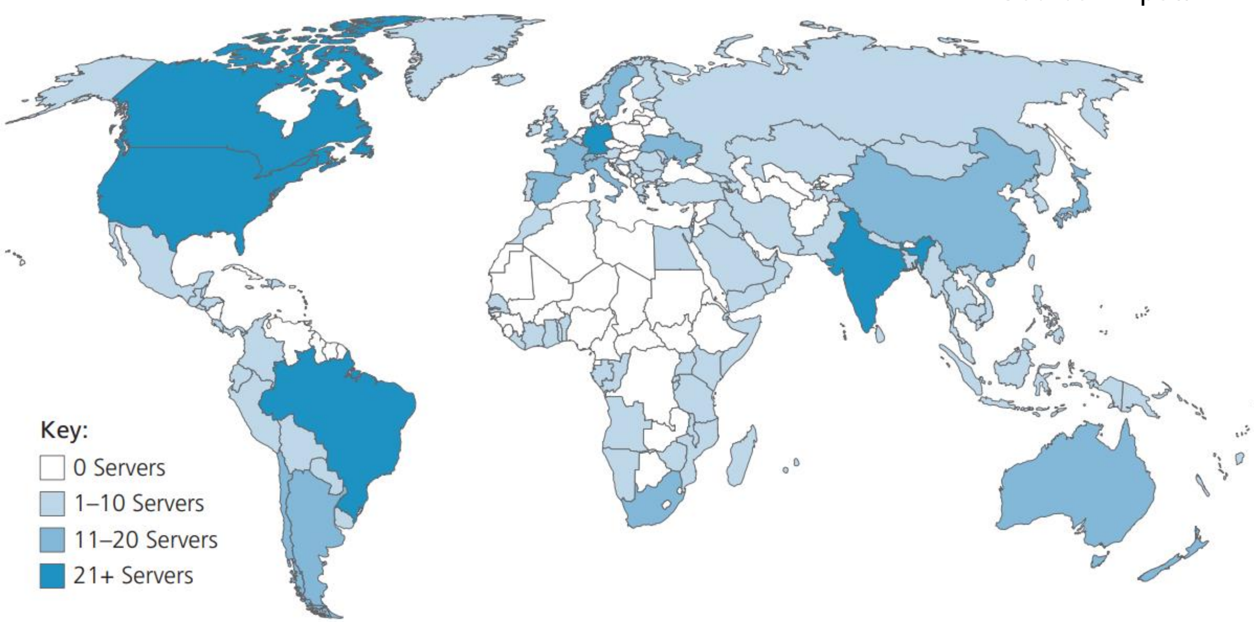
Root DNS Servers

- 13 root DNS servers in the Internet
  - Each server is a cluster of replicated servers
  - More than 1000 root server instances scattered all over the world

List of Root Servers

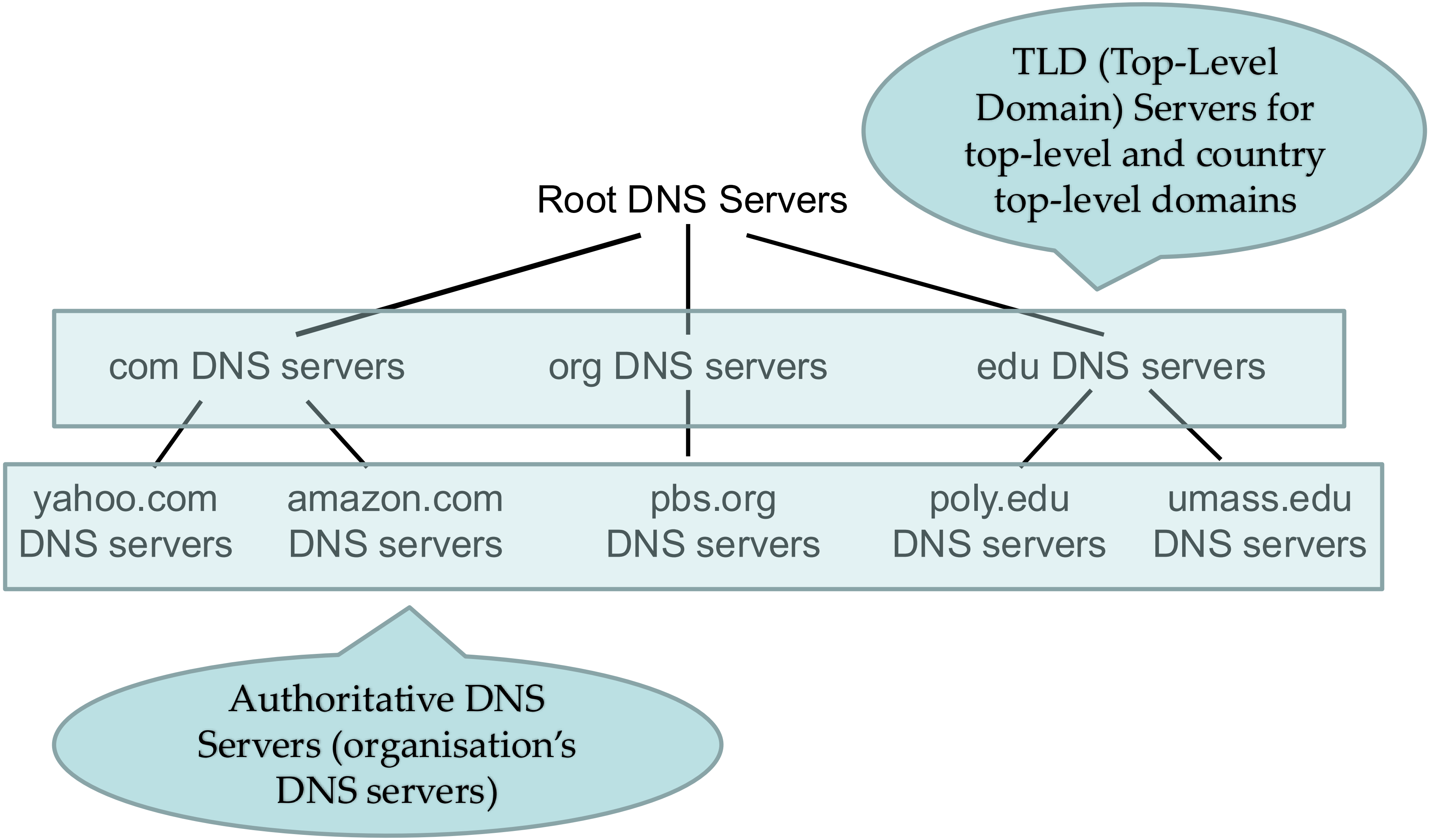
HOSTNAME	IP ADDRESSES	OPERATOR
a.root-servers.net	198.41.0.4, 2001:503:ba3e::2:30	Verisign, Inc.
b.root-servers.net	170.247.170.2, 2801:1b8:10::b	University of Southern California, Information Sciences Institute
c.root-servers.net	192.33.4.12, 2001:500:2::c	Cogent Communications
d.root-servers.net	199.7.91.13, 2001:500:2d::d	University of Maryland
e.root-servers.net	192.203.230.10, 2001:500:a8::e	NASA (Ames Research Center)
f.root-servers.net	192.5.5.241, 2001:500:2f::f	Internet Systems Consortium, Inc.
g.root-servers.net	192.112.36.4, 2001:500:12::d0d	US Department of Defense (NIC)
h.root-servers.net	198.97.190.53, 2001:500:1::53	US Army (Research Lab)
i.root-servers.net	192.36.148.17, 2001:7fe::53	Netnod
j.root-servers.net	192.58.128.30, 2001:503:c27::2:30	Verisign, Inc.
k.root-servers.net	193.0.14.129, 2001:7fd::1	RIPE NCC
l.root-servers.net	199.7.83.42, 2001:500:9f::42	ICANN
m.root-servers.net	202.12.27.33, 2001:dc3::35	WIDE Project

Source: https://www.iana.org/domains/root/servers



Source: James F. Kurose & Keith W. Ross, Computer networking: a top-down approach featuring the Internet, , 8th edition, 2021, Figure 2.18

TLD DNS Servers and Authoritative DNS Servers



Source: https://www.iana.org/domains/root/servers

Local DNS Servers

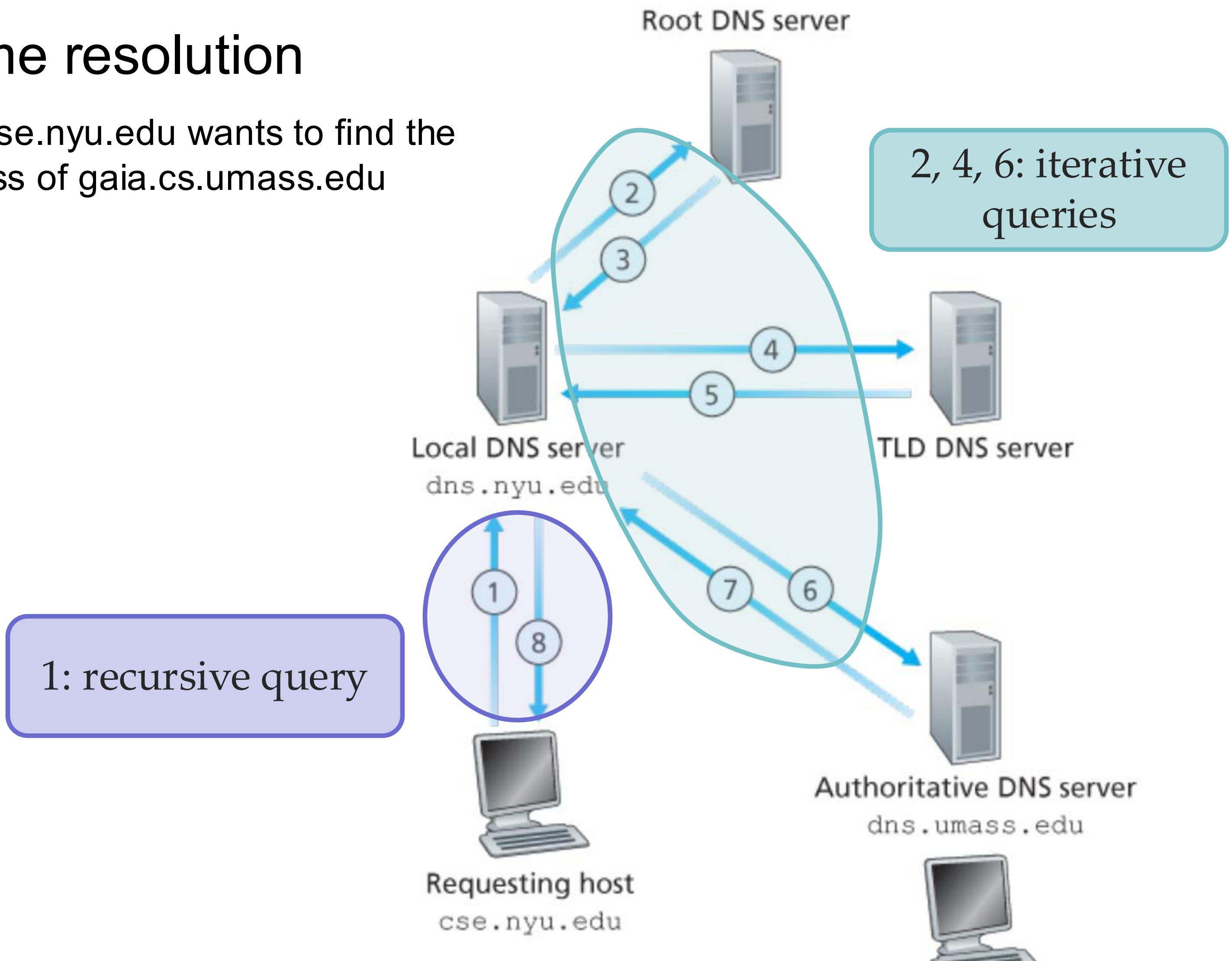
- A local DNS server does not strictly belong to the hierarchy (but is central to the DNS architecture).

- Each ISP (residential or institutional) has a local DNS server (a default name server).

- When a host makes a DNS query, the query is sent to its local DNS server which forwards the query into the hierarchy.

Interaction of the various DNS servers

- DNS name resolution
  - Host at cse.nyu.edu wants to find the IP address of gaia.cs.umass.edu

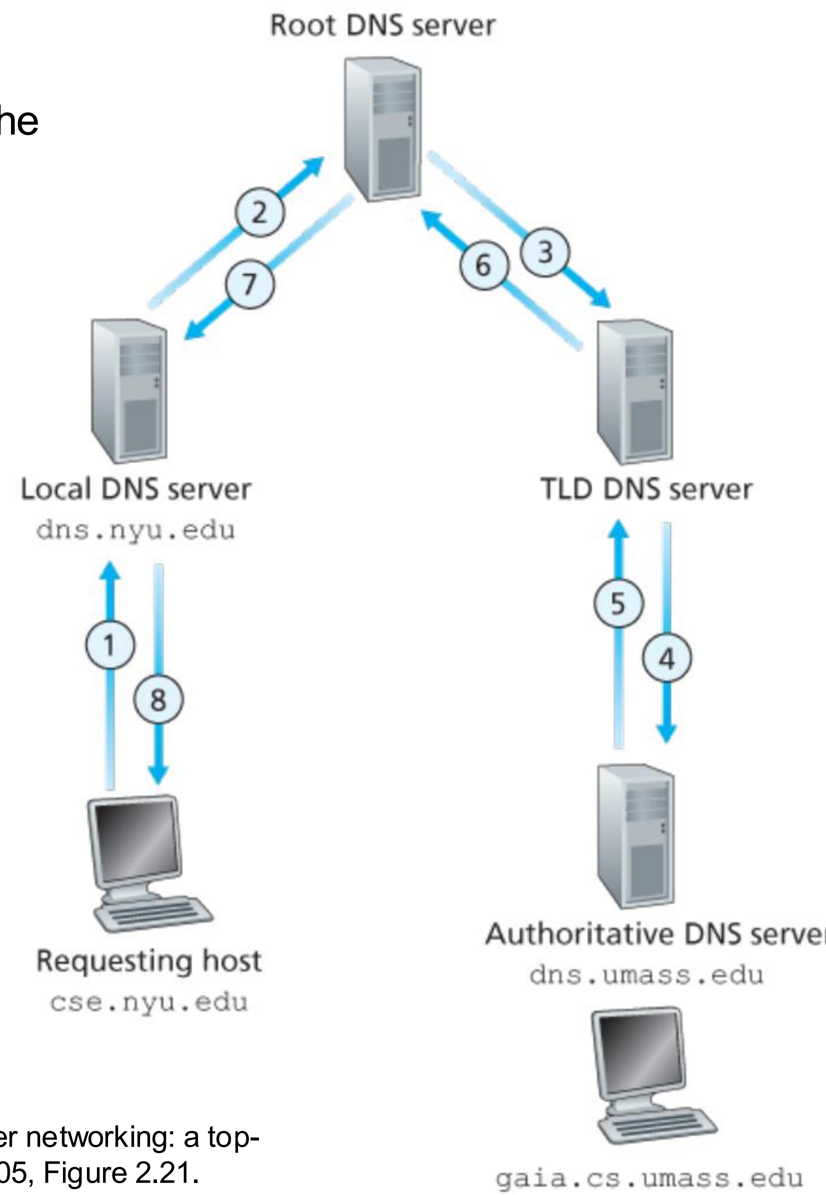


Source: James F. Kurose & Keith W. Ross, Computer networking: a top-down approach featuring the Internet, 3rd edition, 2005, Figure 2.20.

Recursive Queries in DNS

- DNS name resolution
  - Host at cse.nyu.edu wants to find the IP address of gaia.cs.umass.edu

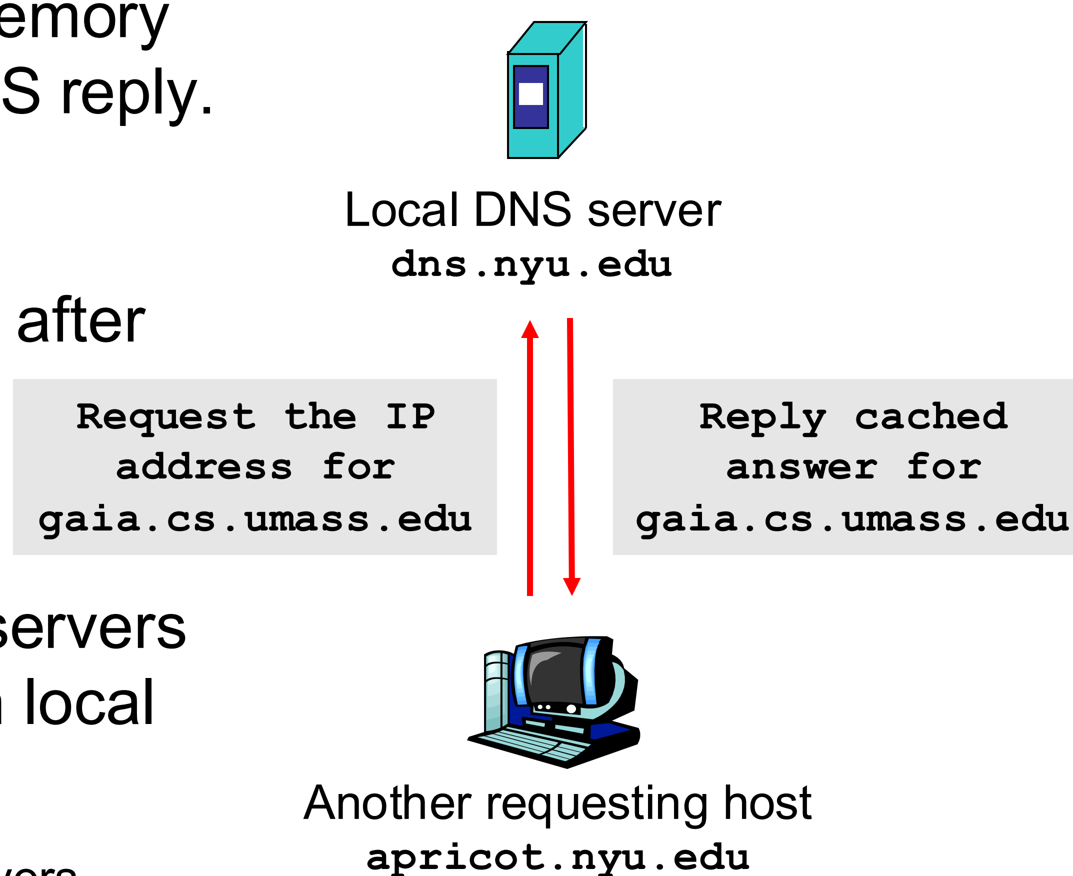
Put burden of name resolution on the contacted name server



Source: James F. Kurose & Keith W. Ross, Computer networking: a top-down approach featuring the Internet, 3rd edition, 2005, Figure 2.21.

DNS Caching

- A DNS server can cache the mapping in its local memory when it receives a DNS reply.
- Cache entries timeout after some time.
- IP addresses of TLD servers are typically cached in local name servers.
  - Bypass the root DNS servers



DNS records

DNS: distributed db storing resource records (RR)

RR format: `(name, value, type, ttl)`

Type=A

name is a hostname
value is the IP address

Type=NS

- **`name`** is a domain (e.g. foo.com)
- **`value`** is the hostname of authoritative name server that knows how to get the IP addrs. for hosts in the domain.

Type=CNAME

name is alias name for some "canonical" (the real) hostname
www.ibm.com is really servereast.backup2.ibm.com
value is canonical name

Type=MX

value is canonical name of a mail server associated with an alias hostname **name**

There are other types: AAAA, PTR, SRV, TXT, etc.

A toy example

name	value	type	TTL
canterbury.ac.nz	web.canterbury.ac.nz	CNAME	XXXX
www.canterbury.ac.nz	web.canterbury.ac.nz	CNAME	XXXX
web.canterbury.ac.nz	132.181.106.9	A	XXXX
canterbury.ac.nz	itdns.canterbury.ac.nz	NS	XXXX
canterbury.ac.nz	mail.canterbury.ac.nz	MX	XXXX
www.google.co.nz	172.217.167.99	A	86400
itdns.canterbury.ac.nz	132.181.2.225	A	XXXX
mail.canterbury.ac.nz	132.181.3.225	A	XXXX

DNS protocol, messages

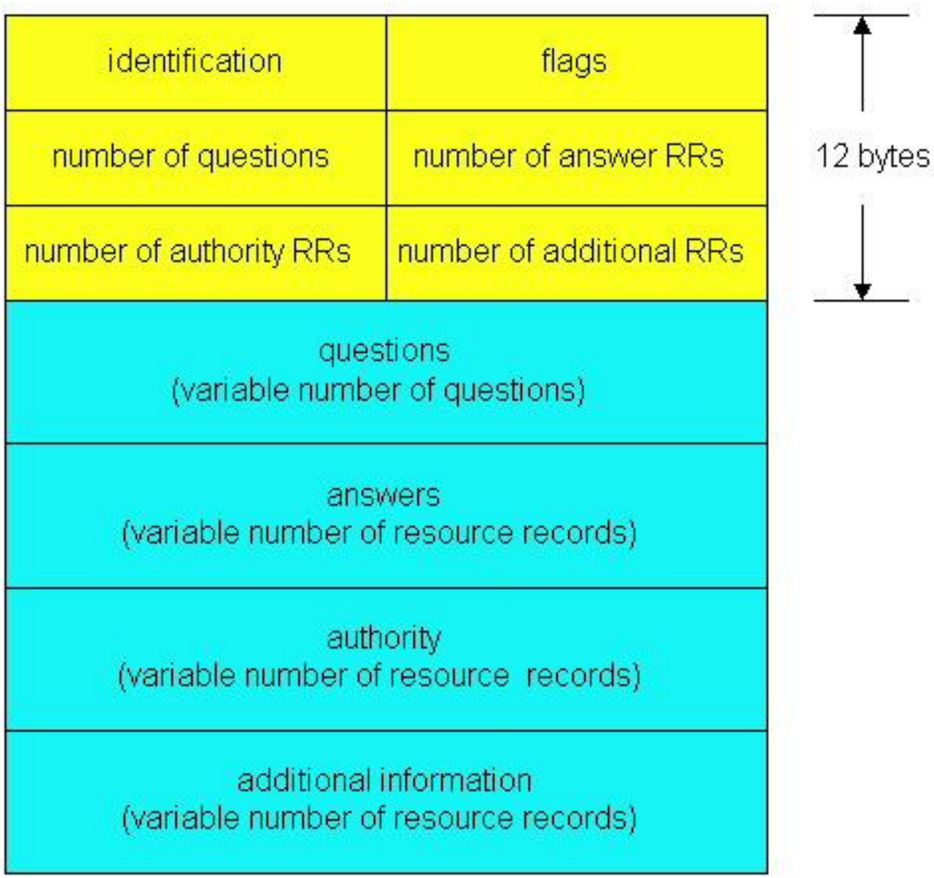
DNS protocol : *query* and *reply* messages, both with same *message format*

msg header

identification: 16 bit # for query, reply to query uses same #

flags:

- Query (0) or reply (1)
- recursion desired (in a query)
- recursion available (in a reply)
- reply is authoritative (in a reply)



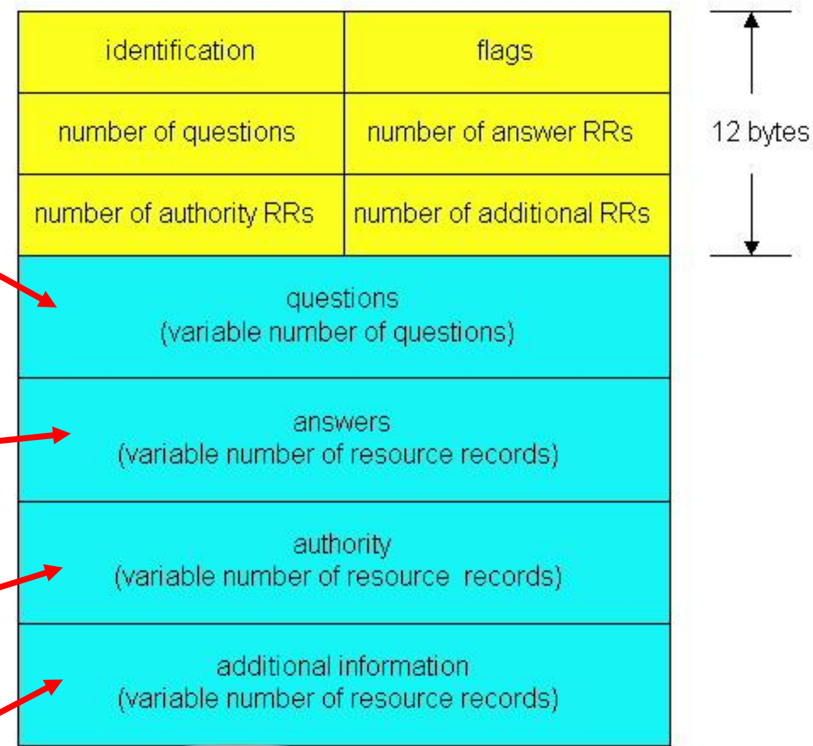
DNS protocol, messages

Name, type fields for a query
(Name, Type)
e.g., (ibm.com, CNAME)

RRs in reponse to query
(Type, Value, TTL)
(CNAME, serv.bckup.ibm.com,5)

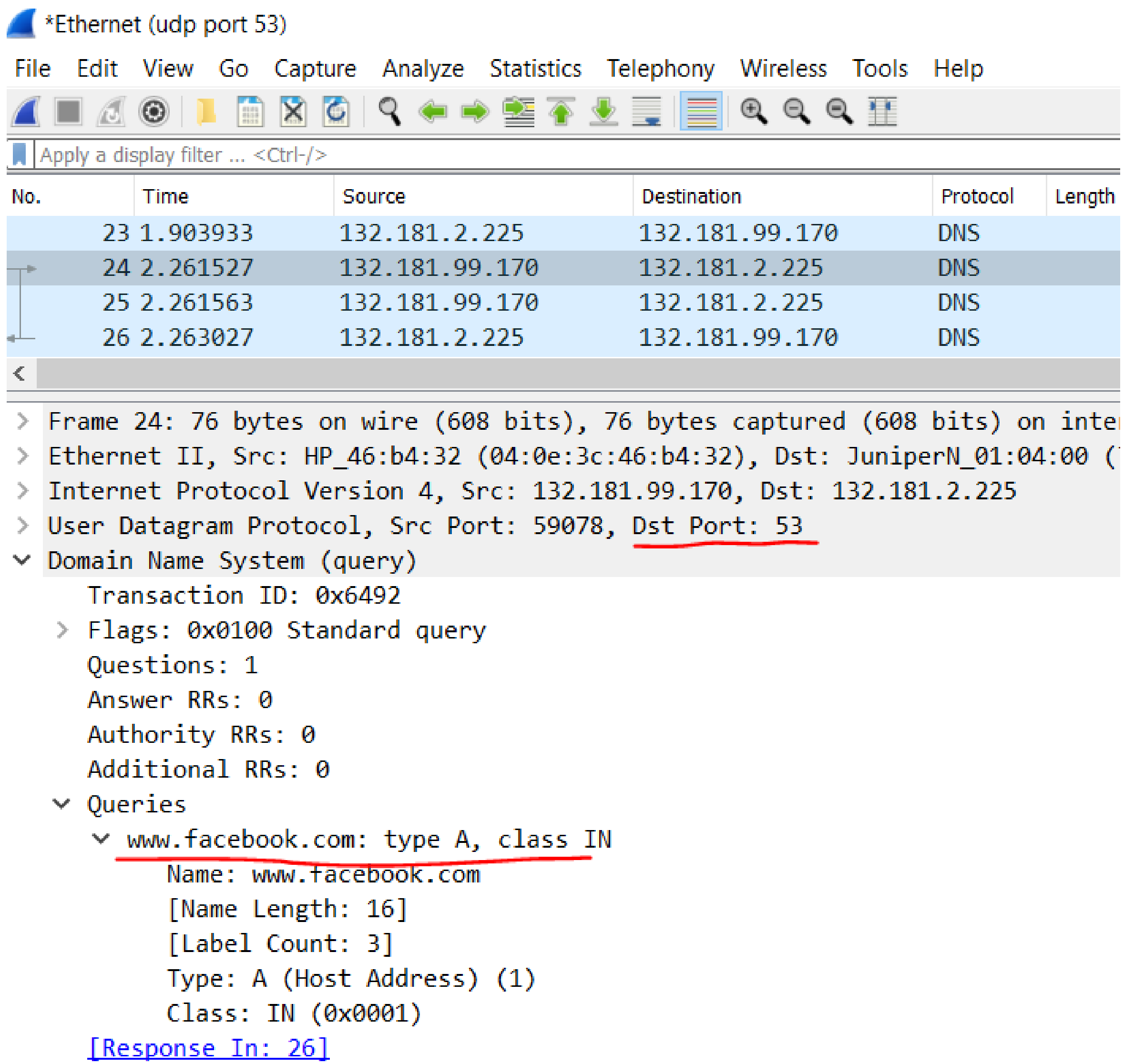
records for authoritative servers

additional “helpful” info that may be used
e.g., (serv.bckup.ibm.com, 254.24.54.42,A)



Multiple RRs possible for the same name!

A Query Example



No.	Time	Source	Destination	Protocol	Length
23	1.903933	132.181.2.225	132.181.99.170	DNS	
24	2.261527	132.181.99.170	132.181.2.225	DNS	
25	2.261563	132.181.99.170	132.181.2.225	DNS	
26	2.263027	132.181.2.225	132.181.99.170	DNS	

```
> Frame 24: 76 bytes on wire (608 bits), 76 bytes captured (608 bits) on inte
> Ethernet II, Src: HP_46:b4:32 (04:0e:3c:46:b4:32), Dst: JuniperN_01:04:00 (
> Internet Protocol Version 4, Src: 132.181.99.170, Dst: 132.181.2.225
> User Datagram Protocol, Src Port: 59078, Dst Port: 53
v Domain Name System (query)
    Transaction ID: 0x6492
  > Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
  v Queries
    v www.facebook.com: type A, class IN
        Name: www.facebook.com
        [Name Length: 16]
        [Label Count: 3]
        Type: A (Host Address) (1)
        Class: IN (0x0001)
    [Response In: 26]
```

A Response Example

```
24 2.261527    132.181.99.170    132.181.2.225     DNS
25 2.261563    132.181.99.170    132.181.2.225     DNS
26 2.263027    132.181.2.225     132.181.99.170    DNS

> Frame 26: 364 bytes on wire (2912 bits), 364 bytes captured (2912 bits) on interface \Device\NPF_-
> Ethernet II, Src: JuniperN_01:04:00 (78:50:7c:01:04:00), Dst: HP_46:b4:32 (04:0e:3c:46:b4:32)
> Internet Protocol Version 4, Src: 132.181.2.225, Dst: 132.181.99.170
> User Datagram Protocol, Src Port: 53, Dst Port: 59078
v Domain Name System (response)
    Transaction ID: 0x6492
  > Flags: 0x8180 Standard query response, No error
    Questions: 1
    Answer RRs: 2
    Authority RRs: 4
    Additional RRs: 8
  > Queries
  v Answers
    > www.facebook.com: type CNAME, class IN, cname star-mini.c10r.facebook.com
    > star-mini.c10r.facebook.com: type A, class IN, addr 31.13.78.35
  v Authoritative nameservers
    > c10r.facebook.com: type NS, class IN, ns d.ns.c10r.facebook.com
    > c10r.facebook.com: type NS, class IN, ns a.ns.c10r.facebook.com
    > c10r.facebook.com: type NS, class IN, ns c.ns.c10r.facebook.com
    > c10r.facebook.com: type NS, class IN, ns b.ns.c10r.facebook.com
  v Additional records
    > d.ns.c10r.facebook.com: type A, class IN, addr 185.89.219.11
    > b.ns.c10r.facebook.com: type A, class IN, addr 129.134.31.11
    > c.ns.c10r.facebook.com: type A, class IN, addr 185.89.218.11
    > a.ns.c10r.facebook.com: type A, class IN, addr 129.134.30.11
    > d.ns.c10r.facebook.com: type AAAA, class IN, addr 2a03:2880:f1fd:b:face:b00c:0:99
    > b.ns.c10r.facebook.com: type AAAA, class IN, addr 2a03:2880:f0fd:b:face:b00c:0:99
    > c.ns.c10r.facebook.com: type AAAA, class IN, addr 2a03:2880:f1fc:b:face:b00c:0:99
    > a.ns.c10r.facebook.com: type AAAA, class IN, addr 2a03:2880:f0fc:b:face:b00c:0:99
    [Request In: 24]
    [Time: 0.001500000 seconds]
```

References

- James F. Kurose & Keith W. Ross, Computer networking: a top-down approach featuring the Internet, Addison Wesley, 3rd edition, 2005.
  - Chapter 2 Application Layer (2.3, 2.4)
- William Stallings, Data and Computer Communications, Pearson, 10th edition, 2013.
  - Chapter 24 Electronic Mail, DNS, and HTTP (24.1, 24.2)
- Andrew S. Tanenbaum & David J. Wetherall, Computer Networks, Prentice Hall, 5th edition, 2010.
  - Chapter 7 The Application Layer (7.1, 7.2)

Acknowledgements

- Lecture notes are developed based on slides from the following three sources:
  - Dr Mengmeng Ge's lecture notes for COSC264, University of Canterbury, 2024
  - Assoc Prof Dan Kim's lecture notes for COSC264, University of Canterbury, 2017
  - Prof Aleksandar Kuzmanovic's lecture notes for CS340, Northwestern University, 2005, https://users.cs.northwestern.edu/~akuzma/classes/CS340-w05/lecture_notes.htm